

- Legumes, whole grains, seeds and black beans in maximising the amount of protein and fibre in the snack.

Product Brief

Snacking is the favourite pastime of the human population, something every single person engages in, and an opportunity to provide much-needed nutrition that hasn't been exploited as yet. The ideal delivery mechanism of these benchmarks would be a simple snack - something that can be eaten at any time of the day, at work or at home, during commutes as well. It must also be noted that the 21st century has been defined by unparalleled scientific and technical innovation, something that must now transfer to the food industry. The food industry must not begin the process of adaptation to the new ways of life, seeking to ensure their consumers' welfare whilst benefiting from the sales of new products. Through the usage and optimization of existing food technologies, and the development of new, more creative ways of providing flavour, the food industry can become a beacon of healthy practices, shaping the wellbeing of all their consumers. This can very easily be done by creating a snacking solution with the following benchmarks and features:-

Nutrition

- Macronutrient Ratio: ~50-55% carbohydrates, 20-25% protein, 20-25% healthy fats
- High in Protein: 10-15g per serving
- Rich in Fiber: 6-8g per serving
- Low in Calories: 250-350 kcal per serving
- Low in Fat: Maximum of 5g per serving
- Iron: 4-6mg per serving, with vitamin C for better absorption
- Calcium: 100-150mg per serving
- Zinc: 5-10mg per serving
- Vitamins: B-complex, A, C, and D

Key Features

- Base Ingredients: Millets (ragi, bajra), chickpeas, lentils, black beans, oats, quinoa
- Savory Flavors: BBQ, garlic parmesan, masala, chaat
- Natural Sweeteners: Dates, jaggery, honey
- No Artificial Preservatives or Additives
- Natural Preservatives: Vitamin C (ascorbic acid), vitamin E (tocopherols), rosemary extract, citric acid
- Alternative Storage: Vacuum-sealing and nitrogen flushing in airtight, moisture-proof packaging
- Anti-Nutrients Reduced: Soaking, fermenting, or sprouting to minimize phytates and oxalate
- Baked, Air-Fried, Roasted

It is essential that the product features a bright colour scheme, differentiating itself from the other snacks found in regular supermarkets in order to captivate the attention of the target audience. To further hone in the solution's ability to cater to the target group, it must be available in four different flavours, each attracting a different sector of the group. Reusable packaging would be key to attracting first-time consumers as it grants them the opportunity to consume the snack over a period of time, rather than in one sitting, something first-time consumers are unlikely to do.

Criterion B: Developing Ideas

Strand 1: Developing design specifications

Design Specifications	Justification	Testing Method
Aesthetics		
Branding - Clean, modern and minimalistic	Branding A clean, modern and minimalistic aesthetic	Customer Surveys Targeted questionnaires

design with warm-tones of brown, green, and accents of blue and purple. - Able to stand out on retail shelves with boldly placed text like ‘High-Protein’ or ‘High-Fibre’. Packaging - Sustainable, eco-friendly packaging in tones of an earthy brown. - Resealable pouch. - Compact design. - Inclusion of labels such as FSSAI approved, or Organic). - Includes a nutrition label marking each nutrition quantity.	provides a premium and modern brand feel, which helps in creating appeal for the product. Labels like ‘Organic’ and ‘High Protein’ can help attract health conscious customers without the need for expensive marketing campaigns. Packaging Sustainable and eco-friendly packaging options will allow consumers to feel good about their purchase, while the reusable nature of the pouch would allow the product to be consumed over a period of time determined by the customer, providing convenience. A compact design also provides greater amounts of portability.	sent to the target audience would help understand the preferences of this demographic. Expert Analysis Analysing the design, the homogeneity of the design components and the produced aesthetic using the help of a graphic designer will help understand the extent of changes required.
Nutrition		
Macronutrients - Calories (350-400 kcal) - Protein (15g-20g) - Fat (10g-12g) - Carbohydrates (40g-45g) - Fibre (7g-10g) Micronutrients - Iron (3mg-5mg, or 15% to 25% of DV) - Sodium (200mg - 250mg) - Potassium (300mg - 400mg) - Calcium (50mg - 100mg) - Magnesium (5mg - 75mg) Vitamins - A (200-250 IU, 5%-10% of DV) - E (3mg - 5mg, 20% to 30% of DV) <i>NOTE: All nutrient values are for one serving, 100g of Snacktive.</i>	Macronutrients 15g of protein for every 100g of Snacktive consumed will appeal to the large numbers of athletes aged 10-14 years old. High protein quantities will also help bolster growth in the crucial ages of puberty. The additional 10g of fibre helps support digestive processes and improve gut health. This also caters to the significant iron deficiency observed in most Indian cities. Micronutrients The inclusion of almost 25% of the daily recommended iron intake helps combat the iron deficiencies faced by most Indian citizens, particularly female consumers. Sodium, Potassium, and Magnesium are all electrolytes crucial for sustained energy. Vitamins The inclusion of almost 10% DV of Vitamin A supports eye health and a stronger immunity system. Vitamin E helps support skin health and holistic wellness.	Nutritional Analysis Through laboratory testing, the various macronutrient and micronutrient parameters can be determined.
Cost		
Cost of Production - Raw Ingredients (Rs 10 - 12) - Labour (Rs 3-4) - Utilities (Rs 1-2) - Quality Control (Rs 0.5 - 1) - Overheads (Rs 2-3) - TOTAL: Rs 16 - 20 Cost of Packaging - Primary Packaging (Rs 4-5) - Secondary Packaging (Rs 2-4) - TOTAL: Rs 6 - 9	The client will have to incur a fee to purchase the product, which is capped by the MRP listed on Snacktive. To ensure the product serves the majority of the Indian population, prices must be pinned to the lower end of the spectrum. A price of Rs 60 for 1 serving of the product will ensure the product is viewed as ‘premium’ while still allowing the masses of the nation to access it on a weekly basis. Snacktive also falls into the same price category as current health foods such as Yoga Bar, The Whole Truth, and Nourishables, pricing themselves between Rs 50 - 80 for 100g, while being significantly cheaper than imported	Cost Analysis Analysis of the cost of raw ingredients bought in bulk alongside the changing costs of production. Understanding the current costs associated with producing 1 serving of the product will help determine the best possible manufacturing strategy.

Cost of Marketing and Distribution - Advertising (Rs 2 - 4) - Logistics (Rs 3 - 4) - TOTAL: Rs 5 - 8 Maximum Retail Price (MRP/MSRP) - Total Cost + 30% - MRP: Rs 45 - 60 <i>NOTE: All values are in the Indian Rupee, and all costs are for the production of 1</i>	health foods (Rs 100 - 150 per 100g). Additionally, residents of Tier 1 and Tier 2 cities are willing to pay between Rs 50 - 100 for on-the-go nutritious snacks.	
Customer		
Target Audience - Primary: Adolescents aged between 10 and 14 years of age. - Secondary: All age groups - All genders. - Middle to upper class (<i>those with disposable income</i>) to allow for spending on premium health snacks. Psychographics - Athletes and sports enthusiasts interested in the pursuit of healthy, nutritious snacking. - Parents interested in clean and nutritious snack alternatives for their children. - Health conscious citizens who embody a special emphasis towards healthy and clean eating. - Working professionals seeking on-the-go nutrition in a cost effective and good-for-you manner. Geographics - Tier 1 and Tier 2 cities (e.g. Hyderabad, Mumbai, Delhi, Bengaluru, Chennai) - Online platforms	The target audience of primarily children aged 10 - 14 is crucial to the success of future generations due to the widespread influence of unhealthy snacks having a negative impact on the holistic wellbeing of children. Adaptation to healthy, nutritious, yet tasty foods will help newer generations understand that food can be healthy and delicious while also being affordable. Snacktive is able to solve the iron and protein deficiency crisis found all over India, across all genders and by integrating these nutrients into the very core of the snack. Secondly, Snacktive is able to cater to those with disposable incomes, found in Tier 1 and Tier 2 cities, who have grown an affinity towards clean eating, generally a result of an urban lifestyle and influence.	Interview An interview with members of the target audience would help understand the varying perspectives and allow the creators of Snacktive to determine the best possible course of action in terms of branding, flavouring and more. Survey A survey of the target audience is also able to achieve this with remarkable efficiency. It becomes easier to survey a larger demographic to determine the best course of action. Larger numbers of respondents would mean a better understanding of the demographic overall.
Size		
The recommended portion size for the product would be 100g, to maximise sales, and production volume while minimising costs.	- The portion size of 100g would allow the snack to be lightweight to carry around, making it an on-the-go snack. - New buyers would be less hesitant to try a product at a lower quantity to reduce wastage and expenditure. - Significantly increases the amount of packages able to be produced at the Rs 60 MRP, allowing for 1000 package production per day with minimal investment.	Weighing Scale Using a weighing scale would allow the mass of each portion to be determined. Ruler A ruler would allow for the determination of the dimensions of each piece and package.

Safety		
Product Safety and Quality Control - Manufactured in accordance with FSSAI (Food Safety and Standards Authority of India). - Adhere to Good Manufacturing Practices (GMP) - Perform periodic nutrition tests with third-party labs across India. - All allergens within the product to be listed on the packaging. Packaging Safety - To be made from food-grade, non-toxic materials that comply with domestic and international regulations. - Materials used for the pouch will be BPA (Bisphenol A) free. - Feature tamper-proof seals on each package. - Resealable packaging. Shelf-Life and Storage - Designed for long shelf life (>6 months) without refrigeration. - Package must be airtight. - Minimise use of chemical preservatives and depend on natural preservatives like salt, vinegar, or natural antioxidants. - Provide a clear expiration date on each package. Consumer Education - Packaging to feature storage instructions such as ‘Store in cool, dry place’ or ‘away from direct sunlight’. - Disclaimers and warnings to be placed on the packaging, emphasizing the consequences of overconsumption and/or substitution of meals.	Product Safety and Quality Control Complying with FSSAI standards provides consumers with the reassurance they need about the quality of produce, while complying with International GMPs would allow Snacktive to provide safely and ethically produced products to customers. Periodic tests would ensure nutritional values are on par with those listed on the package and allow for tweaks as and when necessary. Packaging Safety Using food-grade, non-toxic materials would help maintain the health of Snacktive’s consumers and ensure the product is not becoming harmful due to external factors. BPA’s have been linked to an increased risk of cancer, avoiding them in the packaging is crucial to creating trust between consumers and the brand. The addition of tamper-proof seals would allow the consumer to verify whether the package has been opened, preventing the consumption of a package that has been tampered with by unwanted entities. Shelf-Life and Storage Chemical based preservatives have been linked to an increased likelihood of cancer, making them unwanted for the general consumer. The provision of a clear and visible expiration date would allow the customer to consume the product within the limited time frame.	Self-Evaluation Through thorough and rigorous self-evaluation, the determination of product safety can be made. Using extensive checklists available online through the Indian Government, many determinations can be made. Additionally, for packaging safety, supply-chain origin can be determined to ensure quality and safe practices. Expert Evaluation Through expert evaluation, a tentative understanding of the shelf-life and storage expectations can be determined.
Materials		
Snacktive - Primarily made of high-protein flours such as chickpea flour, almond flour, oatmeal flour, etc. - All ingredients to be of organic nature, in dried or fresh form. - All ingredients must be non-GMO and additive free. - Any protein additives must be natural and largely unprocessed. - Prioritise the usage of locally sourced ingredients to not only reduce costs but support local agricultural industries.	Snacktive A product that is made from natural, organic, and non-GMO ingredients would help speed up the adoption of the product and promote the holistic wellness of the consumer. In light of recent studies and customer hesitancy towards the purchase of synthetic protein supplements, it would be crucial to provide natural protein supplements within the snack. Packages The most important role of the packaging is to increase the shelf-life of the product while serving as the primary marketing device of Snacktive. In	Self-Evaluation Through comprehensive self-evaluation, the origin of the ingredients used for the making of the product, the materials used to make the packaging and more.

Packages - Completely recyclable or biodegradable. - Create a moisture proof barrier layer to prevent spoilage over long-term storage.	order to do this, the packages must cater to the climate conscious populations and maintain a water and air tight barrier between the outside and inside systems.	
Flavour		
Texture - Light and thin with crunch and elegant snap. - Uniform shape, able to handle transportation. - Bite-sized pieces. Taste - Appeals to the target age group. - Includes a combination of savoury, sweet and tangy notes. - Provides variations in taste (ie. different flavours). - Contends with industry standard flavours like desi masala, or chilli garlic.	Texture A light and thin cracker would help create a snappy crunch, similar to that of existing chips and snack options. A uniform shape would allow for packaging to be streamlined, and the size being limited to bite-sized pieces would aid in the convenience of snacking. Taste The fundamental goal with any health product is to make it likeable because often nutrition comes at the expense of flavour. The market is littered with remnants of companies that have produced incredibly healthy products but have failed at market due to lack of flavour. Especially for the ages of 10 to 14, the flavour component is crucial. A cracker that uses a combination of savoury, sweet and tangy notes with the inclusion of various flavour options is bound to succeed at market.	Client Review Surveys, interviews and questionnaires prepared for the clients (the target age group) will help determine the success of the solution.

Success Criteria

Aspect 1: Aesthetics	
1-2	i. The product aesthetic is not able to appeal to a majority of the primary target consumer, aged 10 to 14. ii. The product packaging does not feature earthy, brown tones, complemented by green and blue accents to create one that reflects holistic wellbeing. iii. The product packaging does not feature important information like 'High Fibre', 'High Protein' and 'High Iron' in bold and easily visible locations. iv. The pouch is not reusable in nature and includes labels like 'FSSAI Approved' or 'Non-GMO'. v. The product looks like an unseasoned cracker, not featuring a crisp snap.
3-4	i. The product aesthetic is able to appeal to a minority of the primary target consumer, aged 10 to 14. ii. The product packaging barely features earthy, brown tones, complemented by green and blue accents to create one that reflects holistic wellbeing. iii. The product packaging features one piece of important information like 'High Fibre', 'High Protein' and 'High Iron' in bold and easily visible locations. iv. The pouch is slightly reusable in nature and includes labels like 'FSSAI Approved' or 'Non-GMO'. v. The product looks like a barely-seasoned cracker, not having a bright color featuring a slightly crisp snap.
5-6	i. The product aesthetic is able to appeal to a majority of the primary target consumer, aged 10 to 14. ii. The product packaging largely features earthy, brown tones, complemented by green and blue accents to create one that reflects holistic wellbeing. iii. The product packaging features some important information like 'High Fibre', 'High Protein' and 'High Iron' in bold and easily visible locations. iv. The pouch is mostly reusable in nature and includes labels like 'FSSAI Approved' or 'Non-GMO'. v. The product looks like a slightly-seasoned cracker, having a somewhat bright color featuring a crisp snap.

7-8	i. The product aesthetic is able to appeal to a large majority of the primary target consumer, aged 10 to 14. ii. The product packaging features earthy, brown tones, complemented by green and blue accents to create one that reflects holistic wellbeing. iii. The product packaging features important information like ‘High Fibre’, ‘High Protein’ and ‘High Iron’ in bold and easily visible locations. iv. The pouch is reusable in nature and includes labels like ‘FSSAI Approved’ or ‘Non-GMO’. v. The product looks like a well-seasoned cracker, having a bright color featuring a crisp snap.
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Aspect 2: Nutrition	
1-2	i. Protein content is between 12g and 14g ii. Fibre content is between 2g and 4g iii. Fat is between 16g and 118g, while carbohydrate content is between 52g and 56g. iv. The calorie count of one serving of this snack is between 400 and 420 kcal.
3-4	i. Protein content is between 14g and 16g ii. Fibre content is between 4g and 6g iii. Fat is between 14g and 16g, while carbohydrate content is between 48g and 52g. iv. The calorie count of one serving of this snack is between 370 and 390 kcal.
5-6	i. Protein content is between 16g and 18g ii. Fibre content is between 6g and 8g iii. Fat is between 12g and 14g, while carbohydrate content is between 45g and 48g. iv. The calorie count of one serving of this snack is between 350 and 370 kcal.
7-8	i. Protein content is between 18g and 20g ii. Fibre content is between 8g and 10g iii. Fat is between 10g and 12g, while carbohydrate content is between 43g and 45g. iv. The calorie count of one serving of this snack is between 330 and 350 kcal.

Aspect 3: Cost	
1-2	i. The total cost to prepare one serving of Snacktive is between INR 21 and INR 24. ii. The total cost of packaging for one serving of Snacktive is INR 10 iii. The estimated cost for marketing one serving of Snacktive is between INR 17 and INR 20. iv. The maximum retail price is between INR 58 and INR 64
3-4	i. The total cost to prepare one serving of Snacktive is between INR 19 and INR 21. ii. The total cost of packaging for one serving of Snacktive is INR 8 iii. The estimated cost for marketing one serving of Snacktive is between INR 15 and INR 17. iv. The maximum retail price is between INR 53 and INR 58
5-6	i. The total cost to prepare one serving of Snacktive is between INR 18 and INR 19. ii. The total cost of packaging for one serving of Snacktive is INR 7 iii. The estimated cost for marketing one serving of Snacktive is between INR 13 and INR 15. iv. The maximum retail price is between INR 50 and INR 53
7-8	i. The total cost to prepare one serving of Snacktive is between INR 16 and INR 18. ii. The total cost of packaging for one serving of Snacktive is INR 6 iii. The estimated cost for marketing one serving of Snacktive is between INR 11 and INR 13. iv. The maximum retail price is between INR 45 and INR 50

Strand 2: Develop feasible design ideas
Chip Shape - Design 1